

Report III

Diagonalization of symmetric tridiagonal matrix (quantum harmonic oscillator problem)

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1. Introduction

In quantum mechanics the behavior of a system is described by the Hamiltonian operator. For one-dimensional systems with a potential $V(x)$, the Hamiltonian takes the form:

$$\hat{H} = -\frac{1}{2} \frac{d^2}{dx^2} + V(x)$$

The eigenvalue problem for this operator can be written as:

$$\hat{H}\psi_k = \varepsilon_k \psi_k$$

Where the ψ_k are eigenfunctions and ε_k are corresponding eigenvalues.

In this project we consider the quantum harmonic oscillator with a potential:

$$V(x) = \frac{x^2}{2}$$

which leads to the Hamiltonian:

$$\hat{H} = -\frac{1}{2} \frac{d^2}{dx^2} + \frac{x^2}{2}$$

The goal of this exercise is to solve this eigenproblem numerically using the finite difference method and compare the obtained results with the analytical solution.

2. Description of the Numerical Task

The computational domain is limited to a finite interval:

$$x \in [-x_{\max}, x_{\max}]$$

A grid of n nodes is introduced with:

$$x_i = -x_{\max} + \Delta \cdot i, \quad i = 0, 1, \dots, n-1$$

where:

$$\Delta = \frac{2x_{\max}}{n-1}$$

The second derivative is approximated using the central finite difference:

$$\frac{d^2\psi}{dx^2} \approx \frac{\psi_{i+1} - 2\psi_i + \psi_{i-1}}{\Delta^2}$$

Substituting this into the Hamiltonian leads to a classical matrix eigenvalue problem:

$$H_d \vec{\psi}^{(k)} = \varepsilon_k \vec{\psi}^{(k)}$$

The matrix H_d is tridiagonal with elements defined on:

- diagonal:

$$h_{i,i} = \frac{1}{\Delta^2} + \frac{x_i^2}{2}$$

- sub-diagonal:

$$h_{i,i\pm 1} = -\frac{1}{2\Delta^2}$$

3. Numerical method

The eigenproblem is solved using diagonalization of a symmetric tridiagonal matrix.

In the program we use the LAPACK routine called *LAPACKE_dstev*

This routine takes:

- diagonal elements d
- sub-diagonal elements e

and returns:

- eigenvalues (stored in d)
- eigenvectors (stored in an array)

Eigenvectors are stored in column-major order:

$$\psi_i^{(k)} = z[i + k \cdot n]$$

Each eigenvector is normalized using the discrete approximation:

$$C_k = \sum_{i=0}^{n-1} |\psi_i^{(k)}|^2 \cdot \Delta$$
$$\vec{\psi}^{(k)} \leftarrow \frac{\vec{\psi}^{(k)}}{\sqrt{C_k}}$$

4. Analytical solution

The harmonic oscillator has an exact solution:

$$\psi_k(x) = H_k(x) e^{-\frac{x^2}{2}}$$

where $H_k(x)$ are Hermite polynomials, scaled with a Gauss function.

The eigenvalues are:

$$\varepsilon_k = k + \frac{1}{2}, \quad k = 1, 2, 3 \dots$$

Hermite polynomials are computed using the recurrence relation:

$$H_k(x) = 2xH_{k-1}(x) - 2(k-1)H_{k-2}(x)$$

with initial conditions:

$$H_0 = 1 \text{ and } H_1 = 2x$$

The exact eigenfunctions are evaluated at grid nodes and normalized using the same discrete formula as numerical solutions.

5. Implementation

The program is written in C++ and uses the LAPACK library.

The main steps of the program are:

1. Defining the parameters:

First calculations were performed for $n = 100$, then repeated for $n = 1000$ with

$$x_{max} = 10$$

2. Construction of the grid:

$$x_i = -x_{max} + \Delta \cdot i$$

3. Construction of a tridiagonal Hamiltonian
4. Diagonalization of the matrix:
usage of a LAPACKE_dstev routine
5. Normalization of numerical eigenvectors
6. Computation of exact eigenfunctions:
recurrence relation and then multiplying by Gaussian factor
7. Normalization of exact eigenvectors
8. Saving results to files for both the:
 - eigenvalues
 - eigenvectors

6. Results

The numerical eigenvalues closely follow the analytical values:

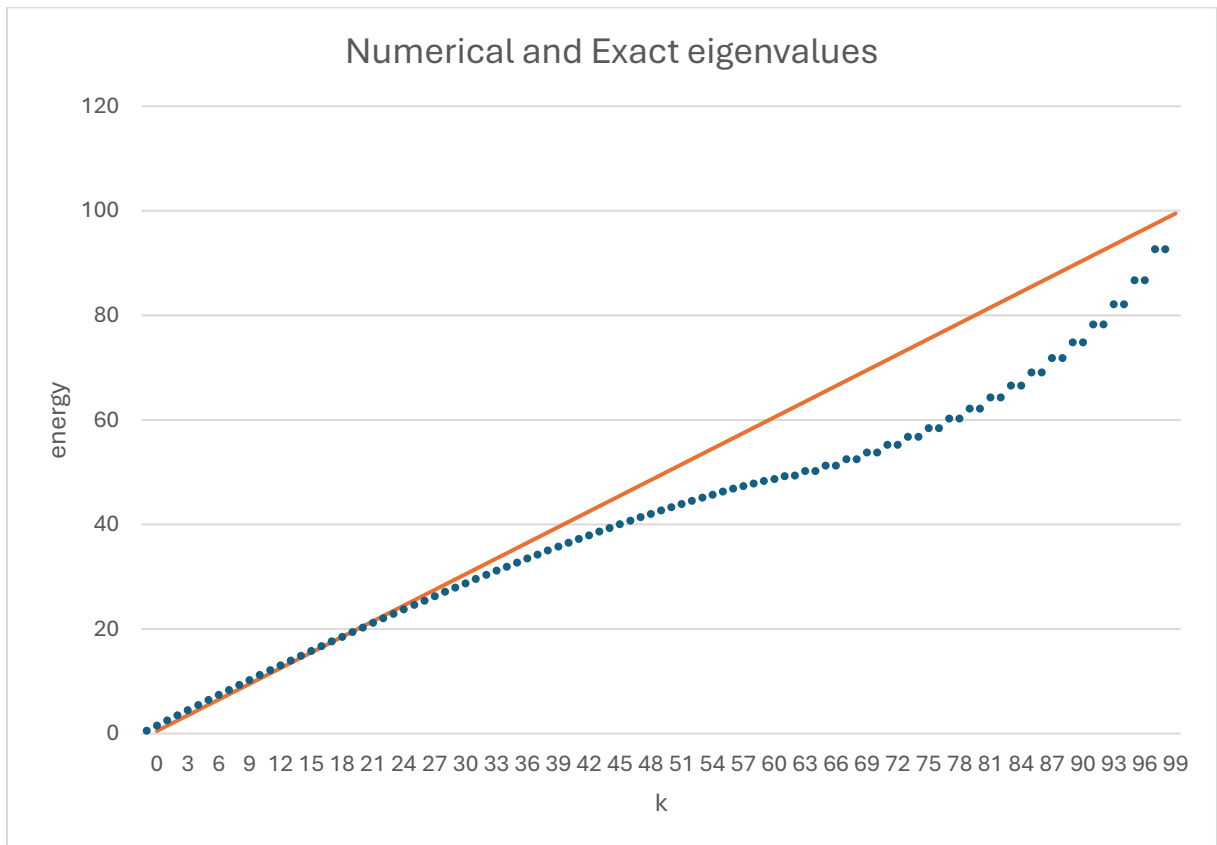
$$\varepsilon_k = k + \frac{1}{2}$$

For small k , the alignment is really good. For higher states, discrepancies tend to increase. However, for $n = 1000$ it showed a better performance up to approximately $k = 50$.

A plot (Fig. 1 and Fig. 3) show:

- exact values forming a straight line
- numerical values deviating at larger k

Figure 1. Comparison of exact eigenenergies (orange line) with eigenenergies obtained from diagonalization (blue dots) for parameters $x_{max} = 10$ and $n = 100$.

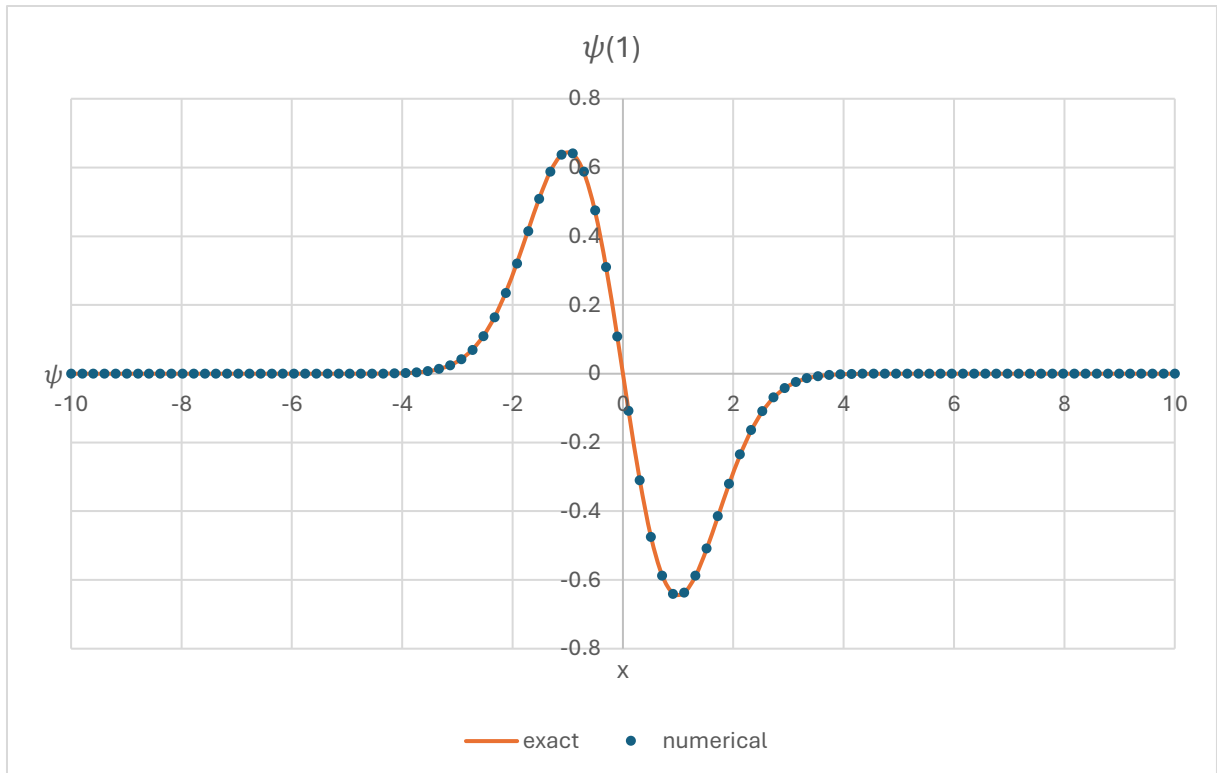


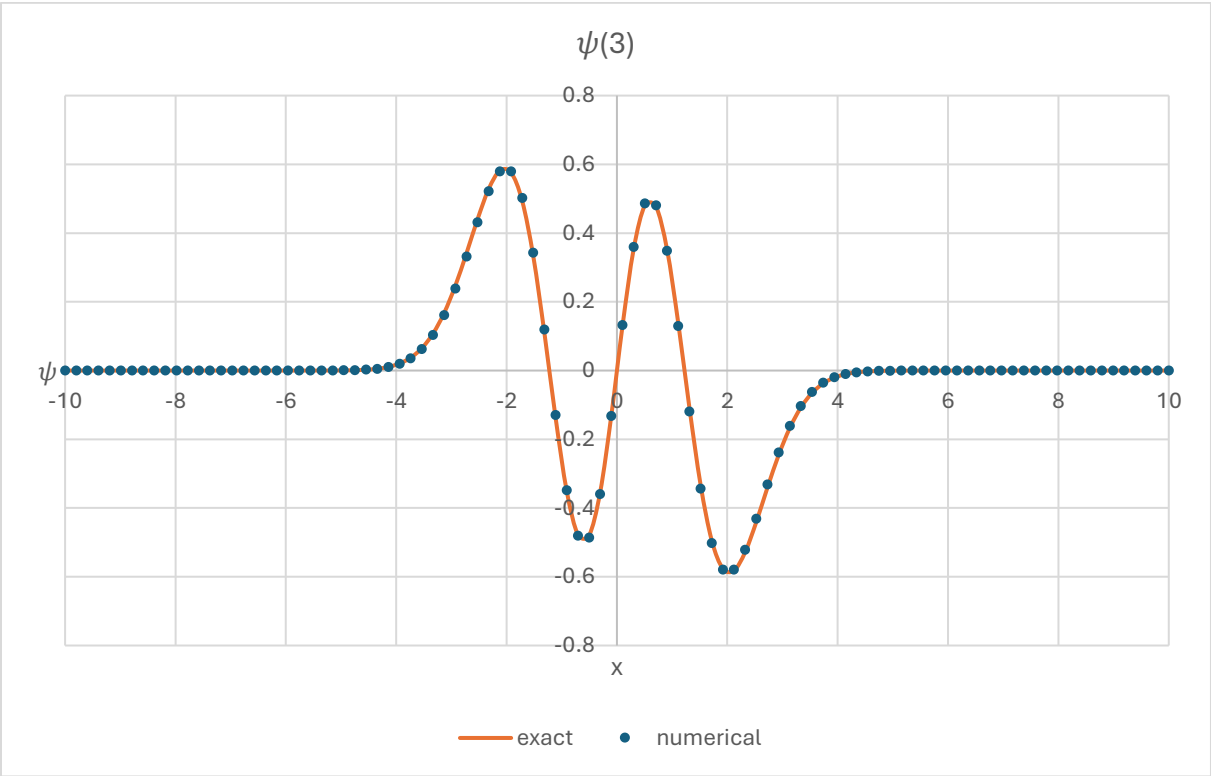
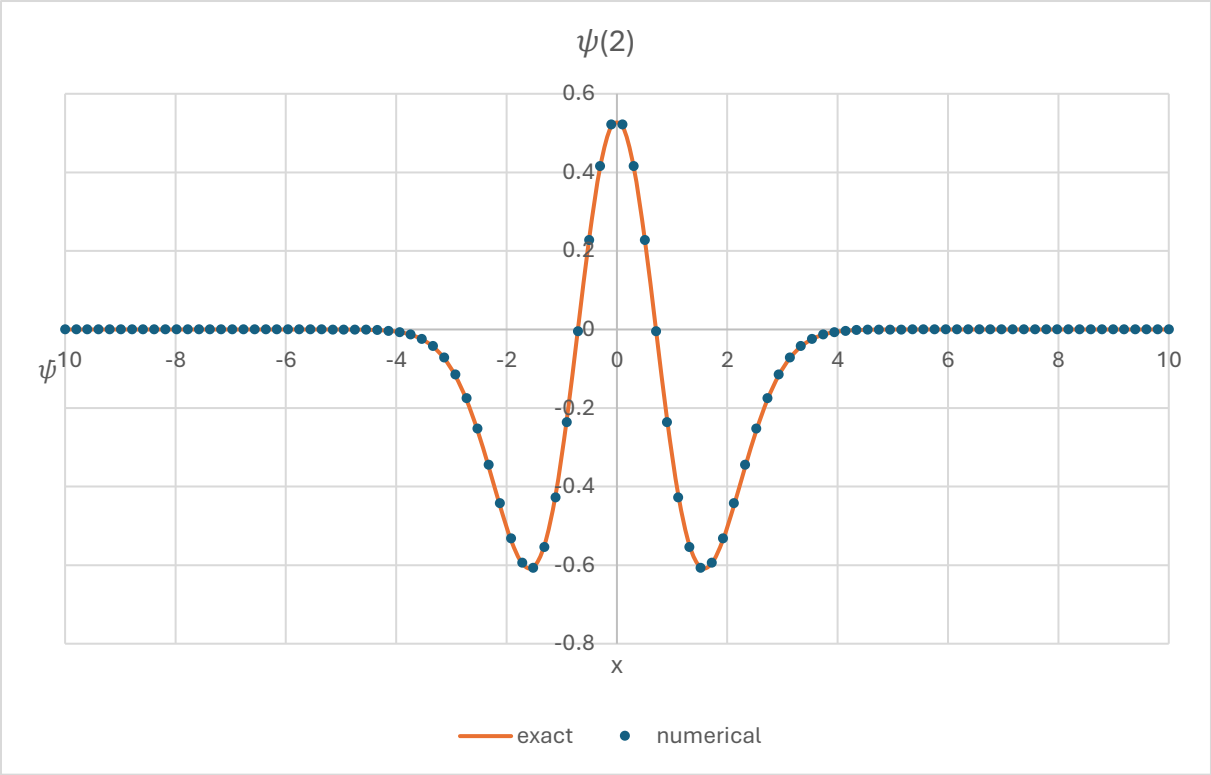
Additionally, the data of eigenvectors was plotted which resulted in graphs of a several wavefunctions for selected states:

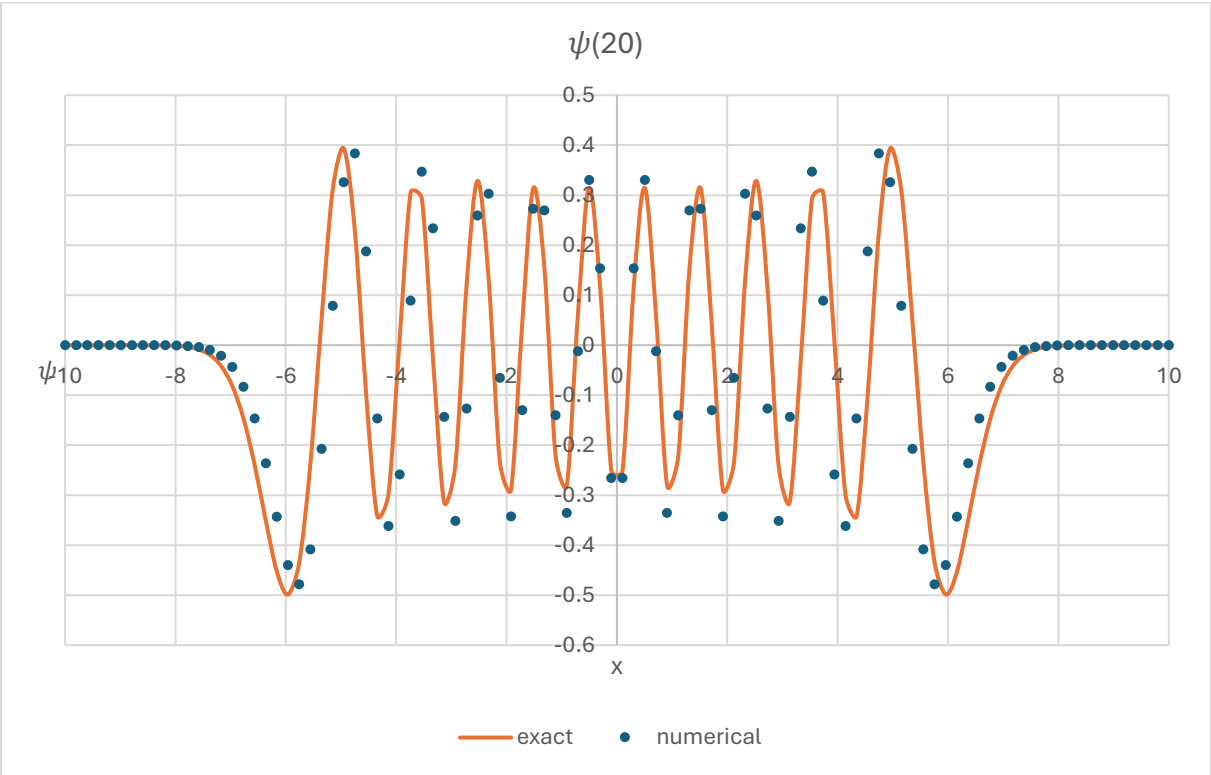
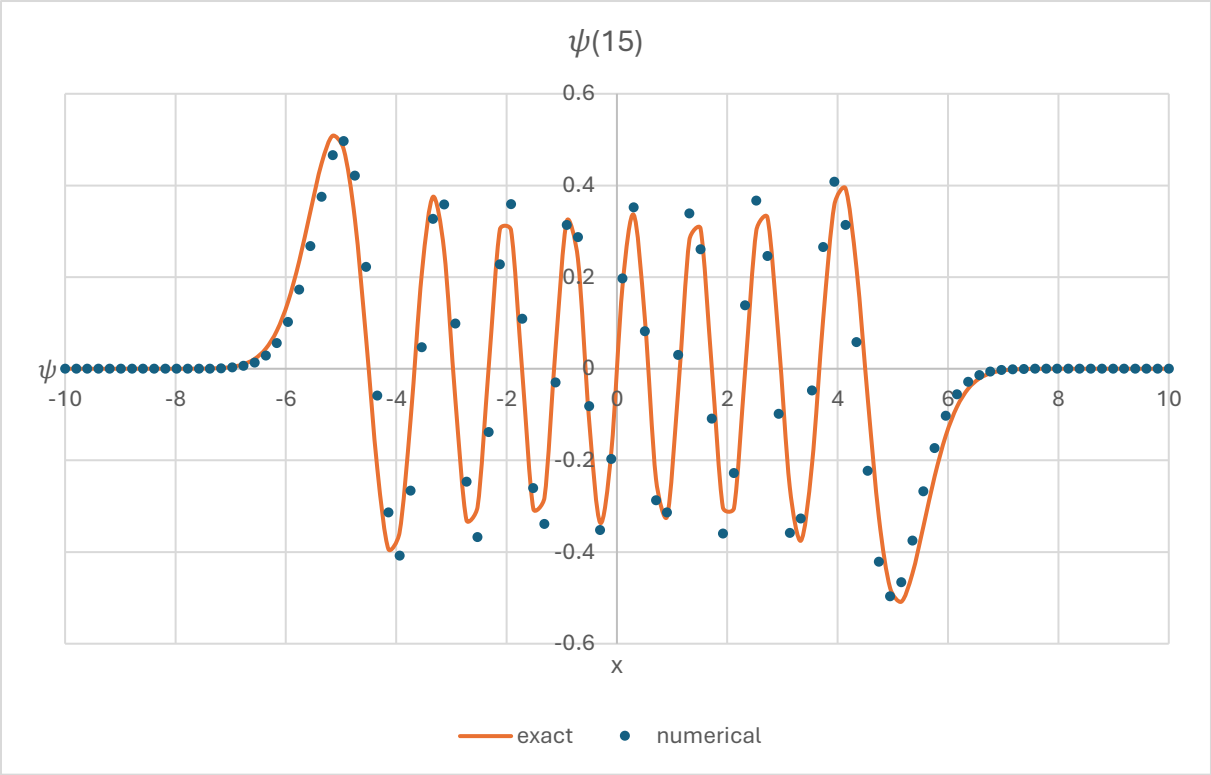
- $\psi^{(1)}$
- $\psi^{(2)}$
- $\psi^{(3)}$
- $\psi^{(15)}$
- $\psi^{(20)}$
- $\psi^{(30)}$

It shows a very good agreement for low states and visible distortion for higher ones.

Figure 2. Example wave functions obtained from diagonalization (blue points) and analytical solutions (orange curve) of harmonic oscillator Hamiltonian for parameters $x_{max} = 10$ and $n = 100$. Indices of states are marked in legends.







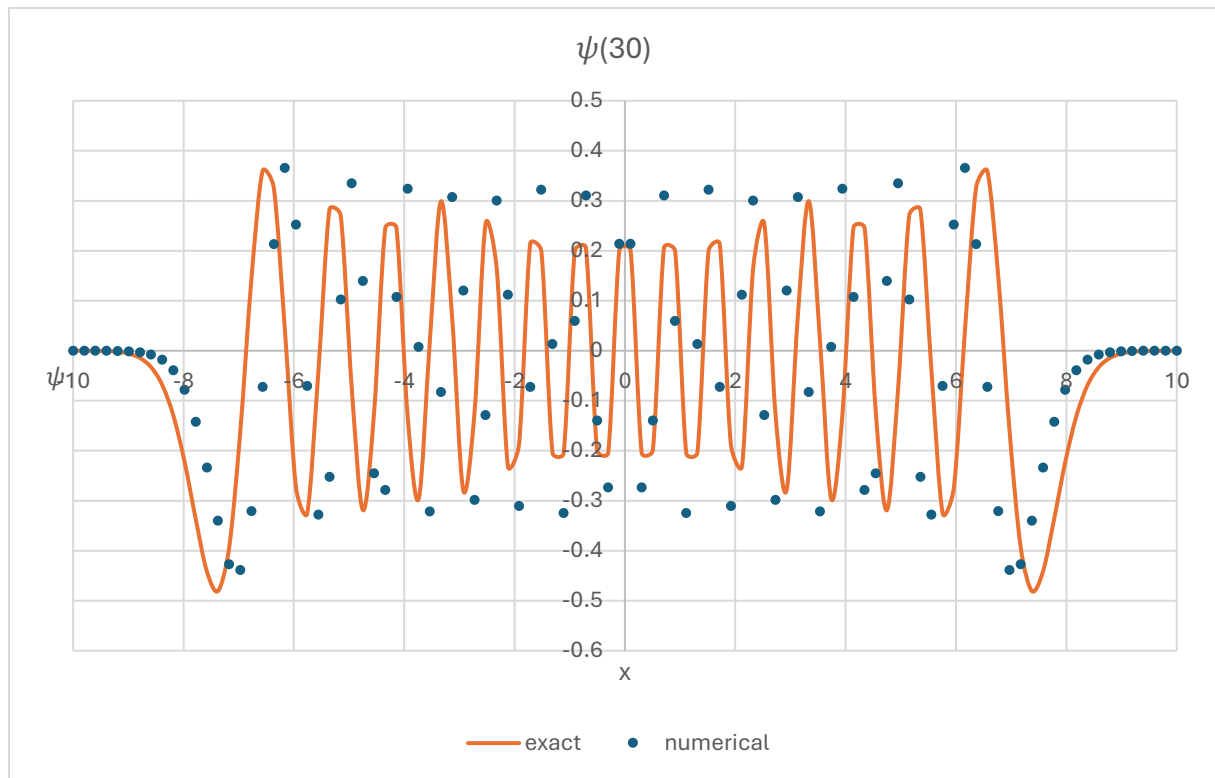


Figure 3. Comparison of exact eigenenergies (orange line) with eigenenergies obtained from diagonalization (blue dots) for parameters $x_{max} = 10$ and $n = 1000$.

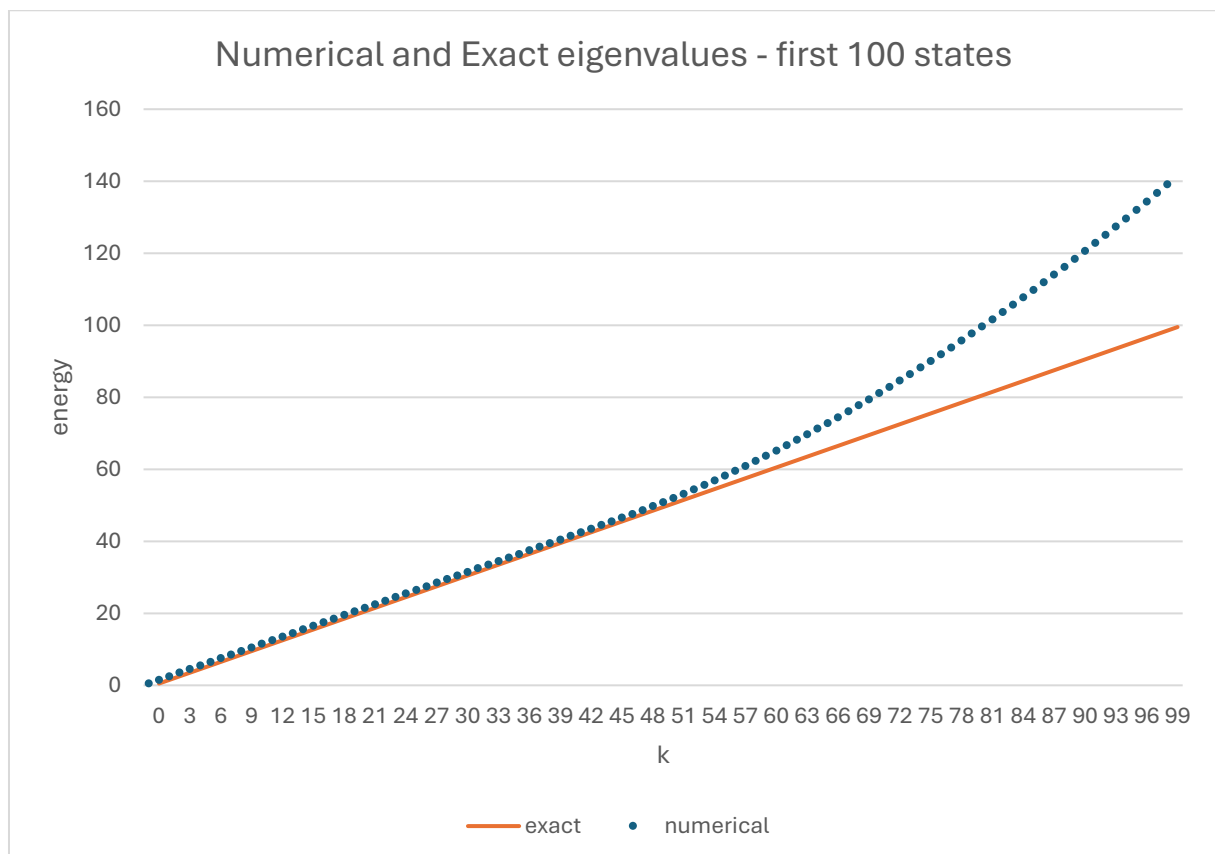
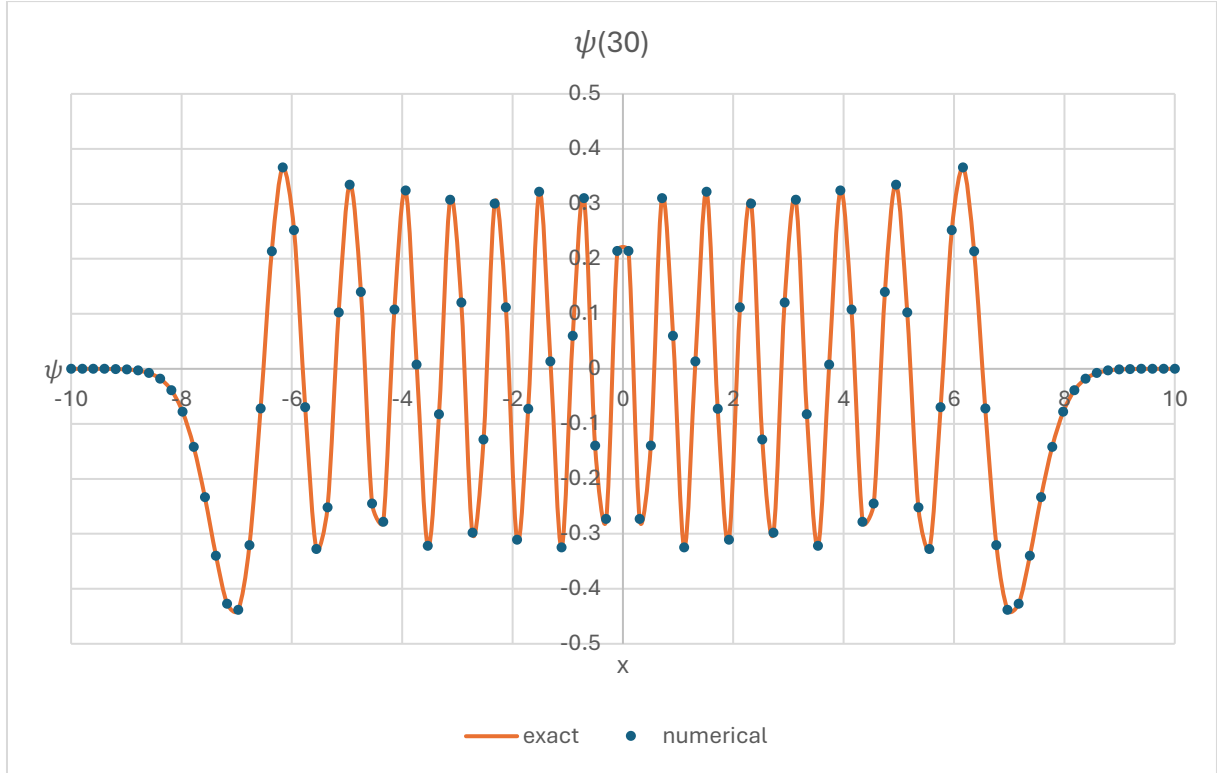


Figure 4. Example of a wave function obtained from diagonalization (blue points) and analytical solutions (orange curve) of harmonic oscillator Hamiltonian for parameters $x_{max} = 10$ and $n = 1000$. The state index is equal to 30.



7. Conclusions

Numerical eigenvalues are not exactly equal to the exact ones, but rather approximate them. The approximation is very good for low-energy states, while for higher-energy states the error increases. This is mainly due to a finite difference approximation of the second derivative, which introduces an $O(\Delta^2)$ error, the limitation of the computational domain, which truncates the wavefunctions.

Eigenvectors corresponding to low quantum numbers (e.g. $k = 1, 2, 3$) are reconstructed very accurately and show excellent precision with the exact solutions. For higher states it deteriorates, as rapidly oscillating wavefunctions are not well captured.

Increasing the number of grid points improves the results. A larger value of n leads to a smaller grid spacing Δ , which provides a better approximation of the second derivative and allows for a more accurate representation of the eigenfunctions. As a result, for small k , both the eigenvalues and eigenvectors are closer to the exact solutions for $n = 1000$, than for $n = 100$.